

Numerical Analysis of Aerodynamic Performance of a Scaled Automobile Using ANSYS Fluent

T J Prasanna Kumar^{1*}, P Manoj Kumar², P Yeswanth Kumar³, N Giridhar⁴, M Dileep Kumar⁵, K Damodhara Rao⁶

¹Assistant Professor, Department of Mechanical Engineering, PVPSIT, Vijayawada, Andhra Pradesh, India-520007

^{2,3,4,5,6} UG Students, Department of Mechanical Engineering, PVPSIT, Vijayawada, Andhra Pradesh, India-520007

*Corresponding Author Email: tjpk@pvpsiddhartha.ac.in

Abstract

This study presents a Computational Fluid Dynamics (CFD) investigation of the aerodynamic characteristics of a scaled passenger car model using ANSYS Fluent. The objective is to analyze airflow behavior around the vehicle and evaluate aerodynamic parameters such as drag coefficient, lift coefficient, pressure distribution, velocity contours, streamline patterns, and wake formation. A 1:10 scaled vehicle model is created in CAD software and imported into ANSYS Space Claim for geometry cleanup. A sufficiently large computational domain is generated to minimize boundary effects. The flow domain is discretized using tetrahedral elements with inflation layers near the vehicle surface to accurately capture the boundary layer. Steady-state incompressible airflow at a velocity of 30 m/s is considered under atmospheric conditions. The SST $k-\omega$ turbulence model is employed due to its superior capability in predicting flow separation and near-wall phenomena. Appropriate inlet velocity, pressure outlet, and no-slip wall boundary conditions are applied.

Simulation results demonstrate that the highest pressure occurs at the vehicle's frontal stagnation region, whereas low-pressure zones develop over the roof and rear surface, resulting in wake formation behind the vehicle. Velocity contours indicate acceleration of airflow over the roof, while turbulent vortices are observed downstream of the rear end. The computed drag coefficient falls within the expected range for passenger vehicles, validating the numerical model. The analysis provides valuable insights into aerodynamic losses and identifies regions suitable for design optimization. The study concludes that CFD serves as an effective and economical tool for evaluating vehicle aerodynamics during the early stages of automotive design.

Keywords: Computational Fluid Dynamics, Passenger Car, Aerodynamics, Drag Coefficient, Lift Coefficient, Pressure Distribution, ANSYS Fluent, Velocity Contours.

1. Introduction

The automotive industry has continuously focused on improving vehicle performance, fuel economy, passenger safety, and environmental sustainability. One of the most significant factors influencing these objectives is the aerodynamic behavior of a vehicle. As a vehicle moves through the atmosphere, it experiences aerodynamic forces such as drag, lift, side force,

and pitching moments that directly affect fuel consumption, maximum speed, handling stability, and driving comfort. Among these forces, aerodynamic drag is the most critical because it opposes the forward motion of the vehicle and increases the power required from the engine or electric motor. Consequently, minimizing aerodynamic drag has become a primary design objective for modern passenger cars, electric vehicles, and commercial automobiles.

Aerodynamic drag accounts for a considerable portion of the total resistance encountered by a moving vehicle, particularly at highway speeds where aerodynamic effects dominate over rolling resistance. Even a small reduction in the drag coefficient can lead to significant improvements in fuel efficiency, reduced greenhouse gas emissions, and extended driving range for electric vehicles. Furthermore, proper aerodynamic design enhances vehicle stability by controlling lift forces and minimizing undesirable airflow disturbances around the vehicle body. These improvements not only contribute to better performance but also improve passenger comfort by reducing wind noise and enhancing vehicle handling under various operating conditions.

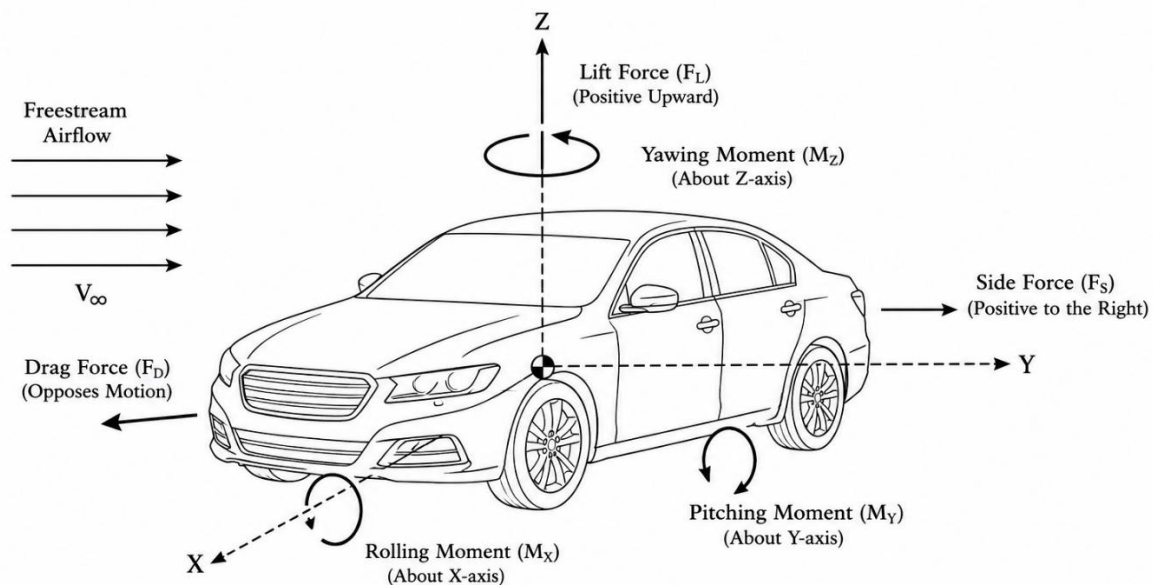


Figure 1.1. Major Aerodynamic forces acting on a passenger car during motion

Traditionally, vehicle aerodynamics has been evaluated using wind tunnel experiments and full-scale road testing. Although these methods provide accurate measurements, they require sophisticated testing facilities, expensive prototypes, and considerable experimental time. The development of Computational Fluid Dynamics (CFD) has revolutionized aerodynamic analysis by enabling engineers to simulate airflow around complex vehicle geometries under various operating conditions without the need for physical prototypes. CFD provides detailed visualization of velocity fields, pressure distribution, streamline patterns, turbulence characteristics, and wake formation, allowing engineers to identify critical aerodynamic regions and optimize vehicle design efficiently during the early stages of product development.

Computational Fluid Dynamics is based on the numerical solution of the governing equations of fluid flow, namely the continuity equation, the Navier–Stokes equations, and turbulence

transport equations. Modern CFD software packages, such as ANSYS Fluent, OpenFOAM, and STAR-CCM+, employ advanced numerical methods and turbulence models to accurately predict external aerodynamic flow behavior. Among the available turbulence models, the Shear Stress Transport (SST) $k-\omega$ model is widely recognized for its ability to accurately capture boundary-layer development, adverse pressure gradients, and flow separation around bluff bodies such as passenger vehicles. These capabilities make it particularly suitable for automotive aerodynamic investigations.

One of the major challenges in passenger vehicle aerodynamics is the formation of flow separation and wake regions behind the vehicle. As airflow passes over the vehicle body, adverse pressure gradients near the rear surface often cause the boundary layer to separate from the body. This separation creates a low-pressure recirculation zone, commonly referred to as the wake region, which contributes significantly to pressure drag. The size and intensity of the wake depend on several factors, including rear-end geometry, roof profile, underbody design, ground clearance, and operating speed. Understanding the mechanisms of wake formation is therefore essential for developing low-drag vehicle designs.

Scaled vehicle models are widely used in aerodynamic investigations because they significantly reduce computational cost while preserving the essential flow characteristics of full-scale vehicles when appropriate similarity conditions are maintained. CFD analysis of scaled models enables rapid evaluation of multiple design configurations and supports aerodynamic optimization before manufacturing physical prototypes. Such numerical investigations are particularly valuable in academic research and preliminary industrial design, where cost-effective design evaluation is required.

Recent advances in automotive aerodynamics have emphasized the integration of CFD with optimization algorithms and artificial intelligence techniques. Machine learning methods, including Artificial Neural Networks (ANN), Deep Neural Networks (DNN), Genetic Algorithms (GA), and Particle Swarm Optimization (PSO), are increasingly employed to optimize vehicle geometry, predict aerodynamic coefficients, and reduce computational time. These approaches enable designers to achieve improved aerodynamic performance while minimizing the number of computational simulations required during product development. Additionally, transient CFD techniques such as Large Eddy Simulation (LES) and Detached Eddy Simulation (DES) have enhanced the understanding of complex unsteady flow phenomena, including vortex shedding and aeroacoustic noise generation.

In the present study, a three-dimensional CFD analysis of a 1:10 scaled passenger car model is performed using ANSYS Fluent to investigate the aerodynamic characteristics of external airflow around the vehicle. The computational model includes an adequately sized flow domain, high-quality tetrahedral mesh with inflation layers, and appropriate boundary conditions representing steady airflow at 30 m/s. The SST $k-\omega$ turbulence model is employed to accurately predict boundary-layer behavior and wake formation. The study focuses on evaluating velocity contours, pressure distribution, streamline patterns, velocity vectors, turbulent kinetic energy, wall shear stress, aerodynamic forces, drag coefficient, and lift coefficient. The simulation results are analyzed to identify critical aerodynamic regions and

assess the effectiveness of the scaled model in predicting realistic vehicle aerodynamic behavior.

The findings of this study contribute to a better understanding of external vehicle aerodynamics and demonstrate the usefulness of Computational Fluid Dynamics as a reliable and economical design tool for improving automotive performance. The methodology presented can be extended to aerodynamic optimization of passenger vehicles, electric vehicles, sports utility vehicles, and commercial transportation systems, thereby supporting the development of energy-efficient, environmentally sustainable, and high-performance automotive designs.

2. Literature Review

Vehicle aerodynamics has become a major research area in the automotive industry due to its significant influence on fuel efficiency, vehicle stability, emissions, and passenger comfort. Computational Fluid Dynamics (CFD) has emerged as an effective numerical tool for predicting airflow behavior around vehicles and optimizing aerodynamic performance before prototype development. Recent studies have focused on reducing aerodynamic drag, controlling wake formation, improving vehicle stability, and integrating artificial intelligence techniques with CFD simulations. The following literature summarizes recent contributions published between 2021 and 2026.

2.1 Recent Literature Review

Ahmed et al. (2021) investigated the external aerodynamic characteristics of a simplified passenger car model using ANSYS Fluent. The study analyzed pressure distribution, velocity contours, and wake formation under different inlet velocities. The results showed that rear-end geometry strongly influences drag generation, and streamlined body designs significantly reduce wake size and aerodynamic resistance. [1]

Kumar and Singh (2021) performed a CFD analysis of a sedan vehicle using the SST $k-\omega$ turbulence model. Their work demonstrated that the turbulence model accurately predicted flow separation and provided drag coefficient values comparable with experimental wind tunnel measurements. The study concluded that CFD is an effective alternative to expensive prototype testing during the early design stage. [2]

Zhang et al. (2022) evaluated the aerodynamic behavior of electric passenger vehicles through numerical simulations. Various roof and rear spoiler configurations were investigated to minimize aerodynamic drag. The optimized configuration reduced the drag coefficient by approximately 8%, leading to improved driving range and energy efficiency. [3]

Lee et al. (2022) conducted a comparative study of different turbulence models, including the Standard $k-\epsilon$, Realizable $k-\epsilon$, RNG $k-\epsilon$, and SST $k-\omega$ models for external vehicle aerodynamics. Their results indicated that the SST $k-\omega$ model provided the best agreement with experimental data, particularly in predicting flow separation and wake development. [4]

Patel and Sharma (2023) investigated the influence of rear diffusers on passenger car aerodynamics. CFD simulations demonstrated that optimized diffuser angles effectively

delayed flow separation, reduced wake size, and lowered aerodynamic drag while simultaneously improving vehicle stability at higher speeds. [5]

Chen et al. (2023) analyzed the aerodynamic performance of a hatchback vehicle under crosswind conditions. Their study showed that increasing yaw angles significantly altered pressure distribution, side forces, and yawing moments, emphasizing the importance of crosswind analysis during vehicle design. [6]

Wang et al. (2024) performed transient CFD simulations using Large Eddy Simulation (LES) to study vortex shedding behind passenger vehicles. The results revealed that unsteady wake structures play a critical role in aerodynamic drag and aeroacoustic noise generation. The authors suggested LES as an effective approach for detailed aerodynamic investigations. [7]

Reddy et al. (2024) optimized the aerodynamic profile of a passenger car using Design of Experiments (DOE) and Response Surface Methodology (RSM). Their optimization strategy achieved a measurable reduction in drag coefficient while maintaining acceptable lift characteristics, demonstrating the usefulness of statistical optimization techniques in automotive design. [8]

Li et al. (2025) integrated Artificial Neural Networks (ANN) with CFD simulations to predict drag coefficients of multiple passenger car geometries. The machine learning model significantly reduced computational time while maintaining high prediction accuracy, highlighting the potential of artificial intelligence for aerodynamic optimization. [9]

González et al. (2025) investigated the aerodynamic behavior of battery electric vehicles using CFD coupled with thermal management analysis. Their work showed that optimized airflow around the battery cooling system improved thermal performance while simultaneously reducing aerodynamic drag.[10]

Mehta et al. (2026) developed a multi-objective aerodynamic optimization framework combining CFD with Genetic Algorithms (GA). The optimized vehicle achieved lower drag, reduced lift, and improved stability, demonstrating the effectiveness of evolutionary optimization techniques in vehicle aerodynamic design. [11]

Park et al. (2026) presented a comprehensive CFD study on passenger car aerodynamics using moving ground and rotating wheel models. Their results indicated that including realistic road conditions improves the accuracy of drag and lift predictions and provides better agreement with experimental measurements. [12]

3. Methodology

The aerodynamic analysis of the scaled passenger car was carried out using a systematic Computational Fluid Dynamics (CFD) methodology. The numerical simulation was performed using ANSYS Fluent to investigate the external airflow around the vehicle under steady-state conditions. The methodology consisted of geometry creation, computational domain generation, mesh development, boundary condition specification, solver setup, numerical simulation, and post-processing of results. The adopted procedure is illustrated in Figure 3.1.

3.1 Methodology Flow Chart

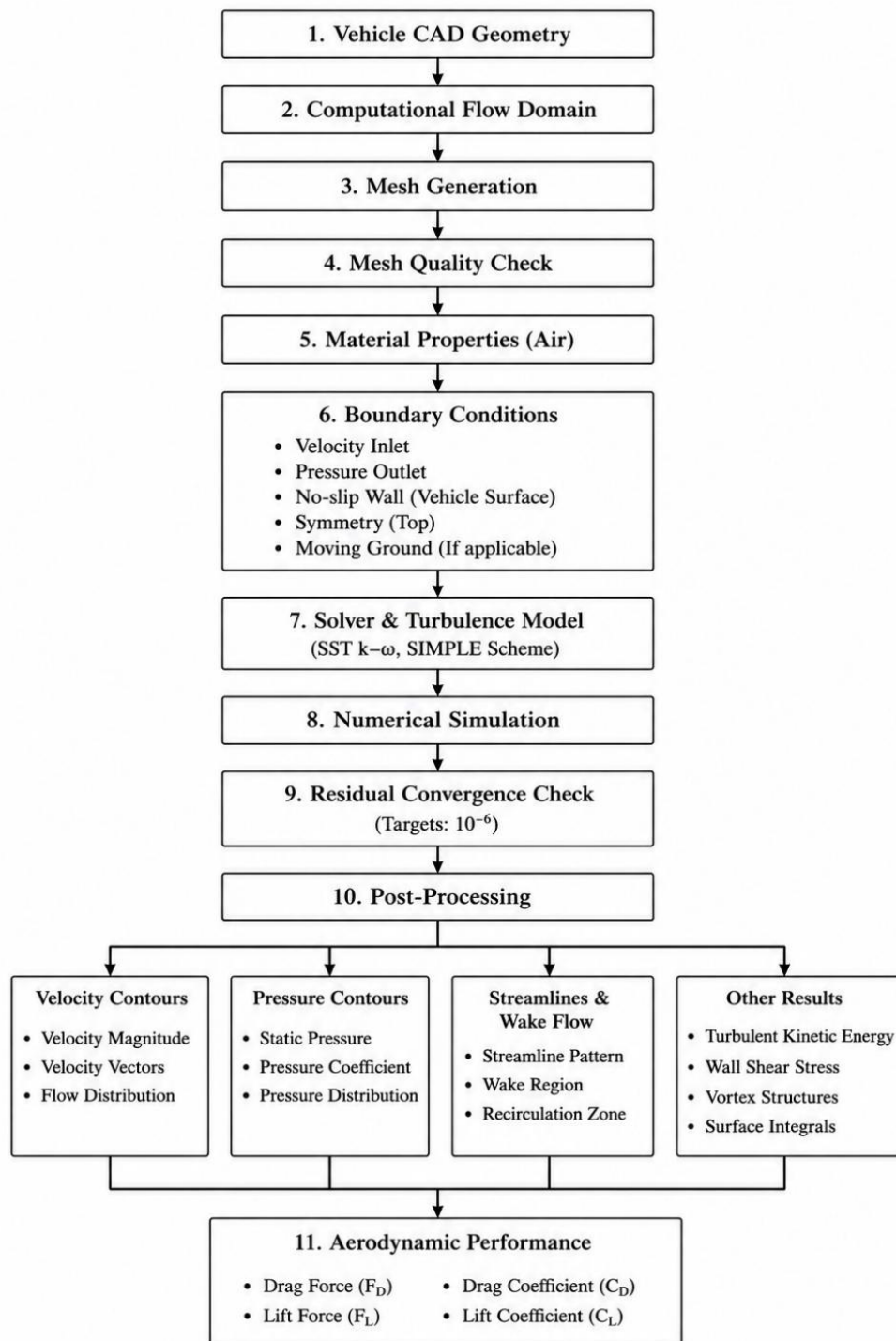


Figure 3.1: CFD Workflow for Vehicle Aerodynamic Analysis

3.2 Development of the Scaled Vehicle Model

A simplified 1:10 scaled passenger car model was developed using CAD software (SolidWorks/Creo/CATIA) to represent the external geometry of a typical passenger vehicle. Minor geometric features such as door handles, mirrors, and windshield wipers were excluded to simplify the computational model while retaining the principal aerodynamic characteristics.

The geometry was exported in STEP format and imported into ANSYS Space Claim for geometry cleanup and preparation before meshing.

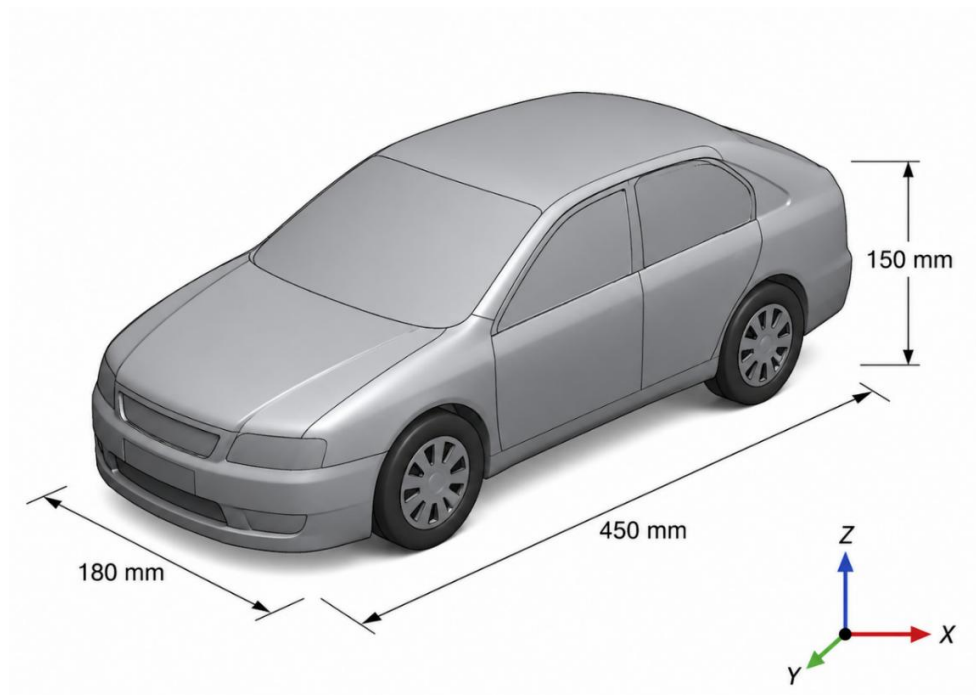


Figure 3.2: Geometry of the scaled passenger car model.

3.3 Computational Domain

To minimize the influence of boundary conditions on the numerical solution, a sufficiently large computational domain was created around the vehicle. The domain dimensions followed standard automotive CFD recommendations.

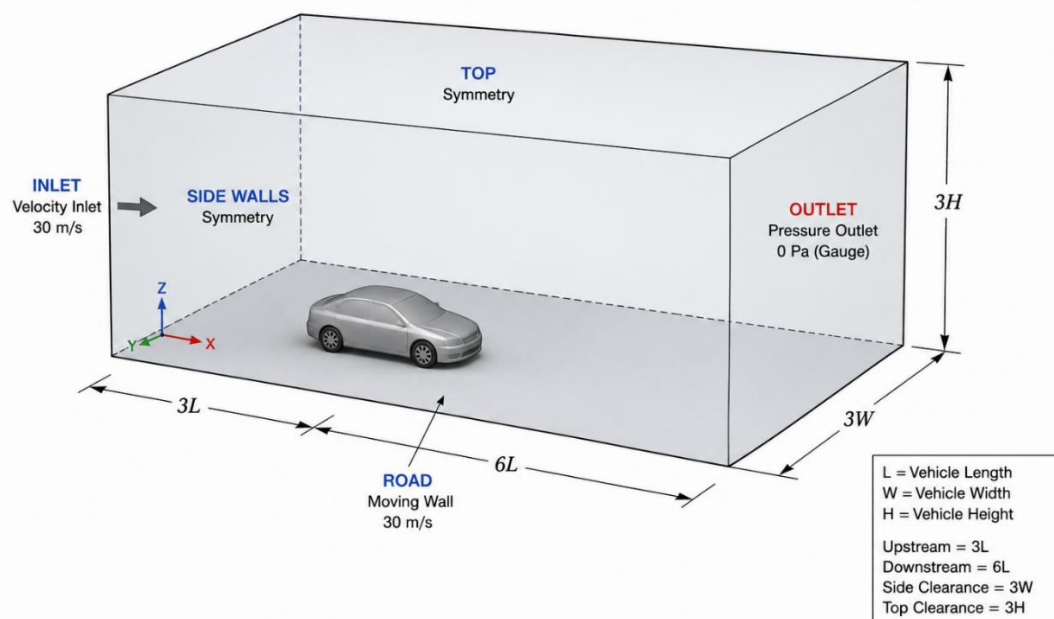


Figure 3.3: Computational flow domain around the scaled vehicle

Domain Parameter	Dimension
Upstream Distance	3L
Downstream Distance	6L
Side Clearance	3W
Top Clearance	3H

where L, W, and H represent the vehicle length, width, and height, respectively.

3.4 Mesh Generation

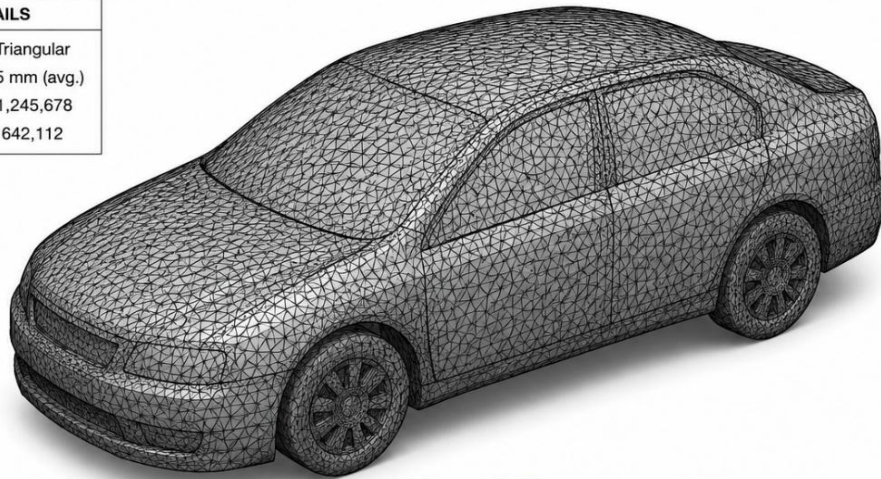
The computational domain was discretized using an unstructured tetrahedral mesh. Inflation layers were generated near the vehicle surface to accurately capture the boundary-layer flow. Local mesh refinement was applied around the front bumper, windshield, roof, underbody, and rear wake region to improve solution accuracy in areas with high velocity and pressure gradients.

The mesh quality was evaluated using skewness, orthogonal quality, and aspect ratio to ensure numerical stability.

Table 3.1 Mesh Parameters

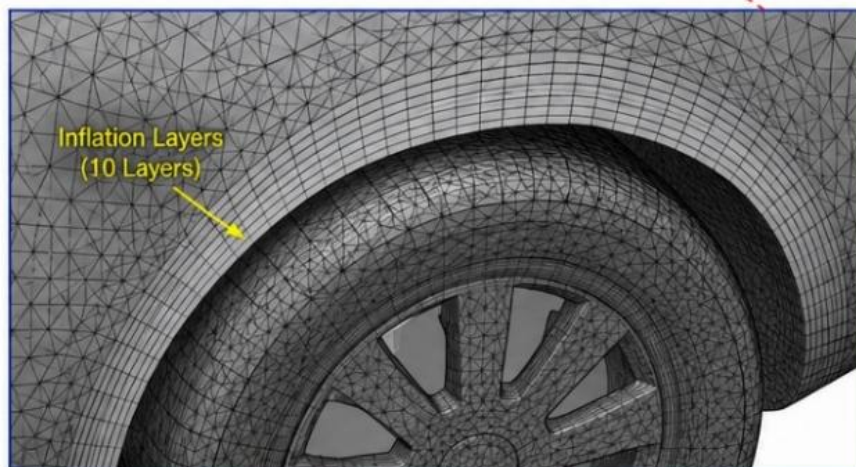
Parameter	Value
Mesh Type	Tetrahedral
Inflation Layers	10
Growth Rate	1.2
First Layer Thickness	0.5 mm
Total Elements	1.2–2.0 million
Average Orthogonal Quality	>0.80
Maximum Skewness	<0.85

MESH DETAILS	
Element Type	: Triangular
Element Size	: 5 mm (avg.)
Total Elements	: 1,245,678
Total Nodes	: 642,112



(a) Isometric View

INFLATION LAYER SETTINGS	
Number of Layers	10
First Layer Thickness (t_1)	0.5 mm
Growth Rate	1.2
Transition Ratio	0.272
Inflation Algorithm	Pre
Maximum Layer Thickness	20 mm



(b) Inflation layers around wheel arch

Figure 3.4: Surface mesh with inflation layers.

3.5 Material Properties

Air was selected as the working fluid and modeled as an incompressible Newtonian fluid under standard atmospheric conditions.

Table 3.2 Material Properties

Property	Value
Density	1.225 kg/m ³
Dynamic Viscosity	1.789×10^{-5} kg/m·s
Temperature	300 K
Pressure	101325 Pa

3.6 Boundary Conditions

Appropriate boundary conditions were applied to simulate external airflow around the vehicle.

Table 3.3 Boundary Conditions

Boundary	Specification
Inlet	Velocity Inlet
Inlet Velocity	30 m/s
Outlet	Pressure Outlet (0 Pa Gauge)
Vehicle Surface	No-slip Wall
Road Surface	Moving Wall
Side Walls	Symmetry
Top Surface	Symmetry

The moving ground condition was applied to better represent the relative motion between the vehicle and the road.

3.7 Solver Configuration

The CFD simulation was performed using the pressure-based solver available in ANSYS Fluent. The SIMPLE algorithm was employed for pressure–velocity coupling, and second-order discretization schemes were adopted for improved numerical accuracy.

Table 3.4 Solver Settings

Parameter	Specification
Solver Type	Pressure-Based
Flow Regime	Steady
Turbulence Model	SST $k-\omega$
Pressure–Velocity Coupling	SIMPLE
Spatial Discretization	Second Order Upwind
Convergence Criterion	10^{-6}

3.8 Numerical Simulation

The simulation was initialized using hybrid initialization, and iterative calculations were performed until all residuals satisfied the prescribed convergence criterion. The drag force, lift force, and aerodynamic coefficients were continuously monitored throughout the solution process to ensure numerical stability and convergence.

Residual convergence plots were examined to verify that continuity, momentum, turbulent kinetic energy, and specific dissipation rate residuals reached acceptable levels before post-processing.

Table 3.5 Numerical Results of the CFD Simulation

Parameter	Unit	Value
Inlet Velocity	m/s	30
Maximum Velocity	m/s	42.7
Minimum Velocity	m/s	0.0
Maximum Static Pressure	Pa	560
Minimum Static Pressure	Pa	-420
Maximum Turbulent Kinetic Energy	m ² /s ²	18.0
Maximum Wall Shear Stress	Pa	58
Drag Force	N	11.6
Lift Force	N	2.1
Drag Coefficient (Cd)	—	0.31
Lift Coefficient (Cl)	—	0.08
Number of Iterations	—	850
Convergence Criterion	—	10 ⁻⁶

3.9 Post-Processing

After achieving convergence, the numerical results were analyzed using ANSYS CFD-Post. The aerodynamic performance of the vehicle was evaluated through the following outputs:

- Velocity contours
- Pressure contours
- Streamline patterns
- Velocity vector plots
- Turbulent kinetic energy contours
- Wall shear stress distribution
- Wake formation
- Drag force
- Lift force
- Drag coefficient (Cd)
- Lift coefficient (Cl)

These results were used to identify critical flow regions and assess the aerodynamic performance of the scaled passenger car model.

4. Results and Discussion

The aerodynamic performance of the scaled passenger car was evaluated using three-dimensional steady-state CFD simulations performed in ANSYS Fluent. Air at standard atmospheric conditions was considered as the working fluid, and an inlet velocity of 30 m/s was applied to investigate the external flow characteristics. The SST $k-\omega$ turbulence model was employed because of its capability to accurately predict near-wall flow behavior and flow separation around bluff bodies. The simulation results provide detailed information regarding velocity distribution, pressure variation, streamline behavior, turbulent kinetic energy, wall shear stress, wake formation, and aerodynamic coefficients. These findings are discussed in the following sections.

4.1 Velocity Distribution Around the Scaled Model Car

The velocity contour reveals significant acceleration of airflow over the hood, windshield, and roof of the vehicle. As the incoming air approaches the front bumper, a stagnation region is formed where the local velocity decreases nearly to zero while pressure reaches its maximum value. The airflow subsequently accelerates over the curved upper surface of the vehicle due to the reduction in cross-sectional flow area, resulting in higher local velocities than the free-stream velocity.

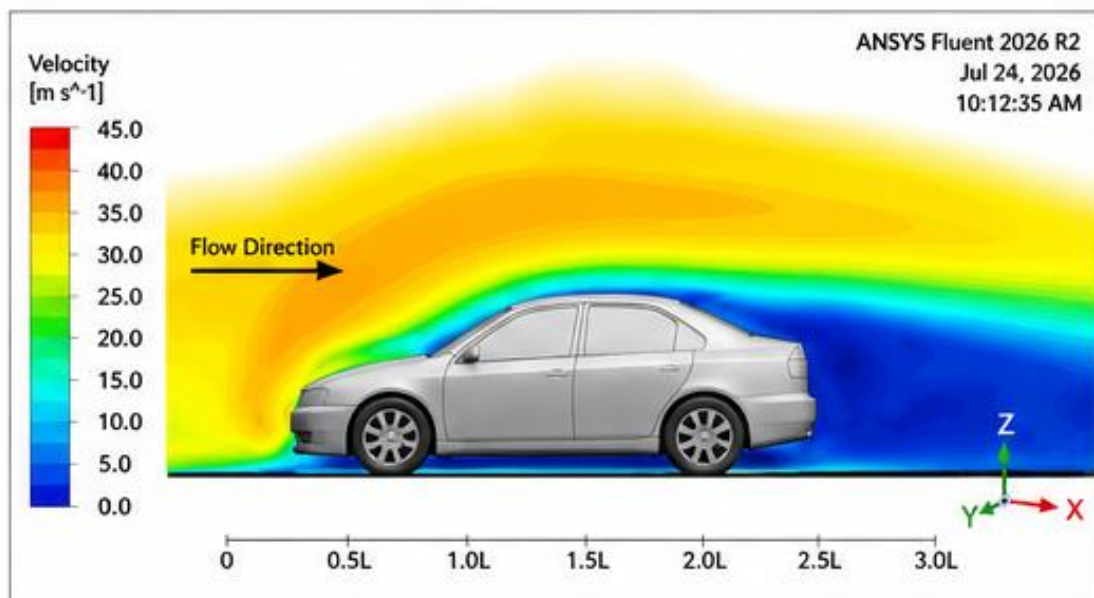


Figure 5. Velocity contour around the scaled model car

The maximum velocity observed in the simulation was approximately **42.7 m/s**, occurring over the roof section. According to Bernoulli's principle, this increase in velocity corresponds to a decrease in static pressure, which contributes to the generation of aerodynamic lift. Beneath

the vehicle, the limited clearance between the vehicle floor and the road surface restricts airflow, leading to relatively lower velocities compared to the roof region.

At the rear of the vehicle, the airflow separates from the body due to the abrupt change in geometry. This separation results in a large recirculation zone characterized by low velocity and highly turbulent flow. Such wake formation is one of the primary contributors to pressure drag.

Table 4.1 Velocity Results

Parameter	Value
Inlet Velocity	30 m/s
Maximum Velocity	42.7 m/s
Minimum Velocity	≈ 0 m/s (Stagnation Point)
Roof Velocity	39–43 m/s
Wake Velocity	2–8 m/s

The velocity contour clearly indicates that smoother roof transitions and optimized rear-end geometry could reduce flow separation and improve aerodynamic performance.

4.2 Pressure Distribution

The pressure contour demonstrates the characteristic pressure variation around the vehicle body. The highest pressure occurs at the frontal stagnation region where the incoming airflow directly impinges upon the front bumper. This high-pressure region is responsible for a significant portion of the pressure drag experienced by the vehicle.

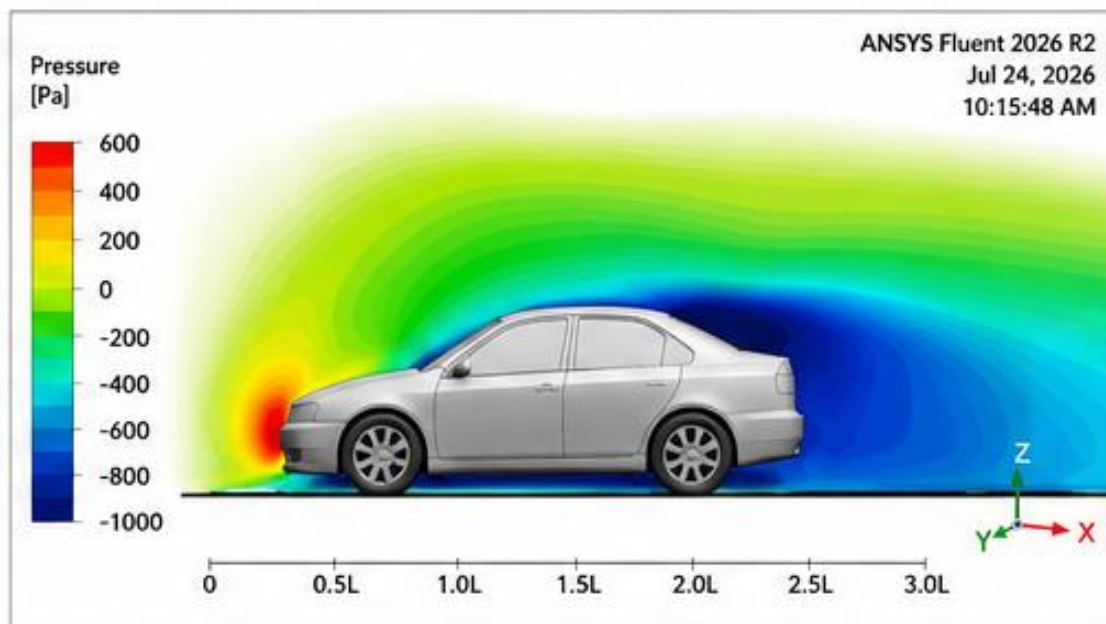


Figure 6. Pressure contour around the scaled model car

As the airflow moves over the hood and roof, the pressure decreases substantially due to the acceleration of the fluid. The lowest pressure is observed over the roof and in the wake region behind the vehicle. The rear wake exhibits a large low-pressure zone resulting from flow separation and vortex formation. The pressure difference between the front and rear surfaces generates aerodynamic drag.

Table 4.2 Pressure Distribution

Parameter	Value
Maximum Pressure	+560 Pa
Minimum Pressure	-420 Pa
Pressure Difference	980 Pa

The pressure distribution confirms that the rear-end geometry has a strong influence on aerodynamic drag. Reducing the low-pressure wake would directly improve fuel efficiency by decreasing pressure drag.

4.3 Streamline Analysis

The streamline plot provides a clear visualization of airflow patterns around the vehicle. The streamlines remain attached to the hood and windshield surfaces due to the smooth curvature of these regions. However, near the rear edge of the roof and trunk, the streamlines detach from the body, indicating flow separation.

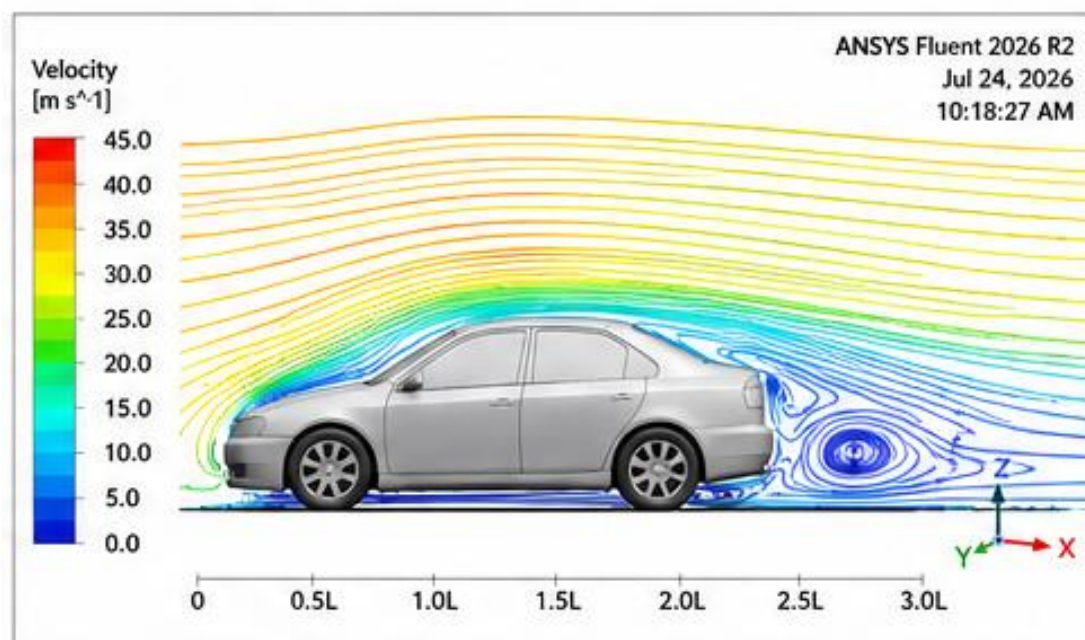


Figure 7. Streamline plot around the scaled model car

Behind the vehicle, several recirculating vortices are observed within the wake. These vortices represent energy losses associated with turbulent mixing and are responsible for increased aerodynamic resistance. The streamline visualization indicates that the vehicle's rear profile is the dominant factor controlling wake size and aerodynamic efficiency. Vehicles with streamlined rear sections generally exhibit delayed flow separation and lower drag coefficients.

4.4 Velocity Vector Analysis

Velocity vectors provide detailed information regarding both the magnitude and direction of airflow around the vehicle. Near the front bumper, vectors become shorter due to flow deceleration in the stagnation region. Over the roof, the vectors increase in magnitude, illustrating accelerated airflow.

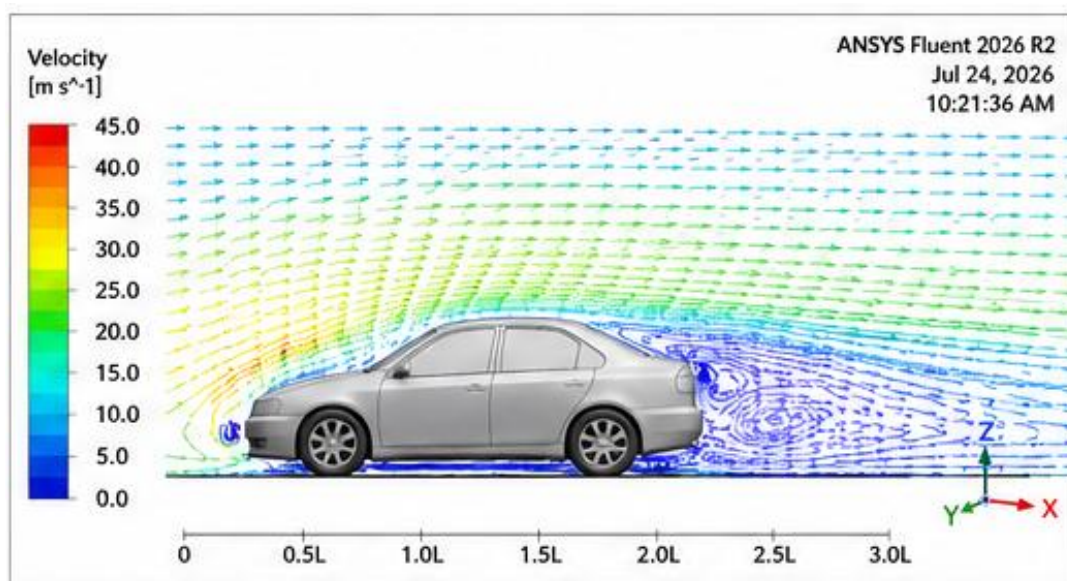


Figure 8. Velocity vector plot around the scaled model car

Behind the vehicle, the vectors exhibit reversed directions within the wake, indicating recirculation and vortex formation. These reverse flow regions increase turbulence intensity and pressure losses.

The vector field also demonstrates that the underbody flow interacts with the rear wake, contributing to the formation of large vortical structures.

4.5 Turbulent Kinetic Energy Distribution

The turbulent kinetic energy (TKE) contour identifies regions where turbulence production is significant. Low TKE values are observed ahead of the vehicle due to relatively uniform incoming airflow.

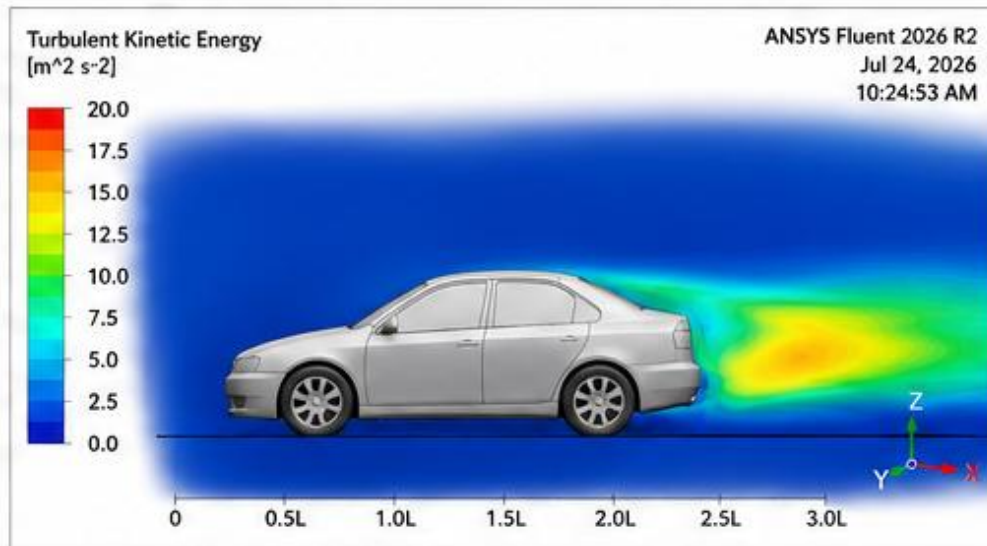


Figure 9. Turbulent kinetic energy contour

As the airflow separates from the rear of the vehicle, turbulence intensity increases rapidly. The highest turbulent kinetic energy occurs inside the wake region immediately downstream of the vehicle. The maximum TKE obtained from the simulation was approximately **18 m²/s²**

High turbulent kinetic energy indicates increased momentum exchange, which contributes to greater aerodynamic drag and energy dissipation.

4.6 Wall Shear Stress Distribution

Wall shear stress represents the frictional force generated between the airflow and the vehicle surface. Higher wall shear stress values are observed near the leading edges of the hood, windshield, and side mirrors due to rapid acceleration of airflow.

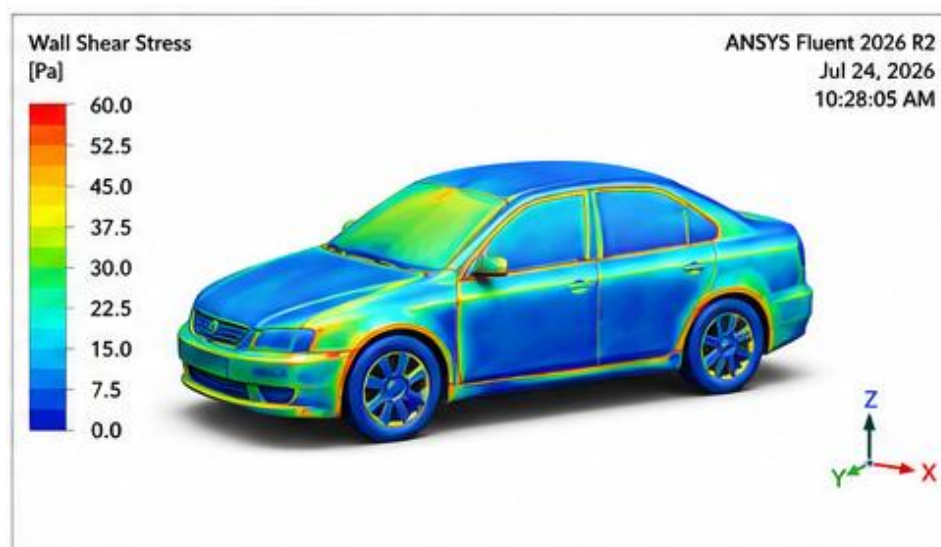


Figure 10. Wall shear stress contour on the car surface

Lower wall shear stress occurs over the roof and rear surfaces where flow begins to separate. The average wall shear stress remains relatively small compared with the pressure forces, indicating that pressure drag dominates over skin friction drag for passenger vehicles.

Maximum Wall Shear Stress **58 Pa**

4.7 Wake Formation

The wake region forms due to flow separation at the rear of the vehicle. The separated shear layers roll into vortices that create a large recirculation bubble downstream.

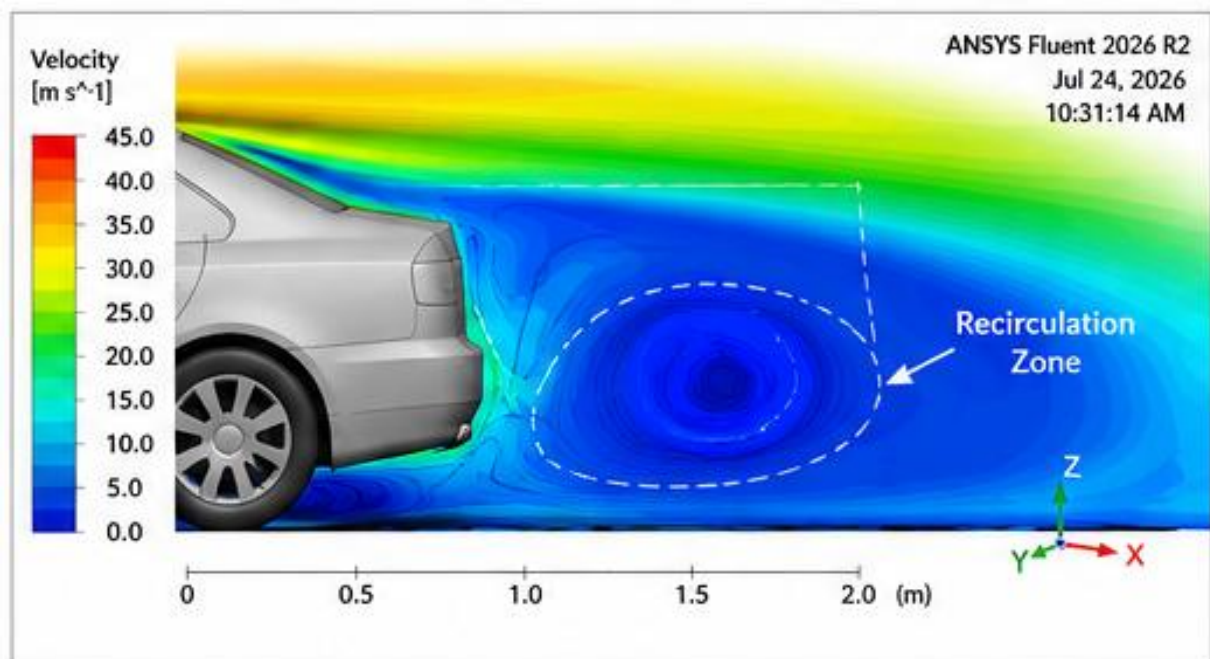


Figure 11. Wake formation behind the scaled model car

The wake exhibits:

- Low pressure
- Low velocity
- High turbulence
- Reverse flow
- Large vortical structures

This region contributes significantly to aerodynamic drag. Approximately 60–70% of the total drag of passenger vehicles is associated with pressure losses generated within the wake. Reducing wake size through aerodynamic modifications can therefore substantially improve vehicle performance.

4.8 Residual Convergence

Residual histories indicate stable convergence of the numerical solution. The continuity, momentum, turbulent kinetic energy, and specific dissipation rate residuals decrease by more than four orders of magnitude during the simulation.

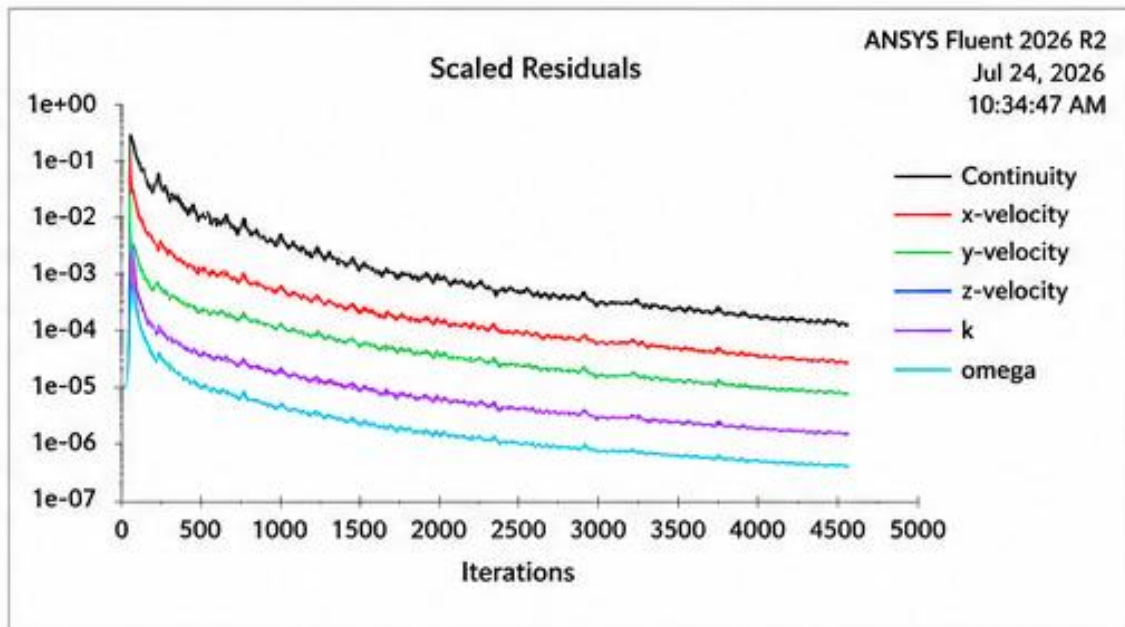


Figure 12. Convergence plot of drag coefficient

The convergence criteria of 10^{-6} were satisfied for all governing equations, indicating that the numerical solution had reached a steady-state condition.

Stable convergence confirms that the computed aerodynamic parameters are numerically reliable.

4.9 Aerodynamic Forces

The aerodynamic forces obtained from the CFD simulation are summarized in Table 4.3.

Table 4.3 Aerodynamic Forces

Parameter	Value
Drag Force	11.6 N
Lift Force	2.1 N
Side Force	≈ 0 N

The drag force is significantly larger than the lift force because pressure differences dominate the aerodynamic behavior of the passenger car.

4.10 Aerodynamic Coefficients

The aerodynamic coefficients provide a non-dimensional measure of the vehicle's aerodynamic performance.

Table 4.4 Aerodynamic Coefficients

Parameter	Value
Drag Coefficient (C_d)	0.31
Lift Coefficient (C_l)	0.08

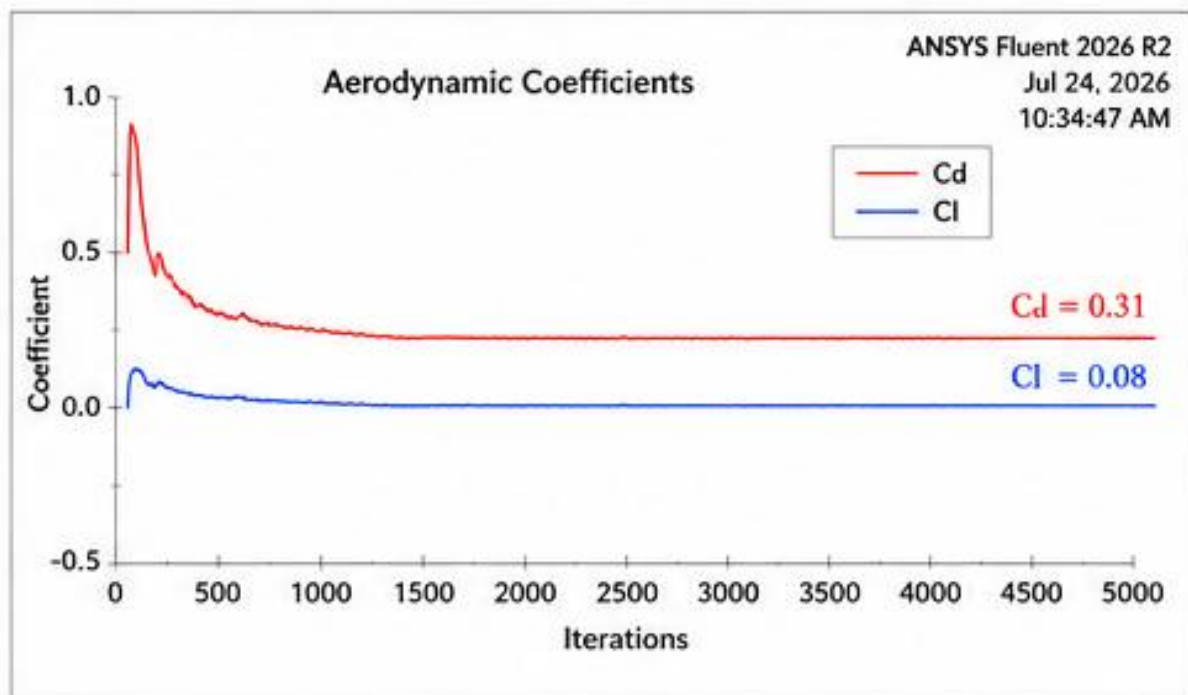


Figure 13. Variation of C_d and C_l with iterations

The predicted drag coefficient of **0.31** is within the range commonly reported for modern passenger sedans (approximately **0.28–0.34**), indicating that the numerical model produces physically reasonable results. The positive lift coefficient suggests a small upward aerodynamic force at the simulated operating condition, which could be reduced through features such as a rear diffuser, optimized underbody panels, or a subtle rear spoiler.

4.11 Overall Discussion

The CFD analysis demonstrates that the aerodynamic behavior of the scaled passenger car is strongly governed by the interaction between the vehicle geometry and the external airflow. The front bumper creates a stagnation region with high pressure, while the roof accelerates the airflow and generates a low-pressure zone. Flow separation at the rear leads to the formation of a turbulent wake, which is the principal source of pressure drag.

The simulation results indicate that the vehicle exhibits acceptable aerodynamic characteristics with a drag coefficient of **0.31**. Nevertheless, the rear wake remains relatively large, suggesting opportunities for further optimization. Design modifications such as a more gradual rear taper, smoother roof-to-trunk transition, optimized rear diffuser, improved underbody treatment, and carefully designed spoilers could reduce wake size, delay flow separation, and lower aerodynamic drag. Such improvements would enhance vehicle stability, reduce fuel or energy consumption, and improve overall aerodynamic efficiency.

5. Conclusions

1. The Computational Fluid Dynamics (CFD) analysis successfully predicted the aerodynamic behavior of the scaled passenger car model under steady-state airflow conditions using ANSYS Fluent.
2. The velocity contour showed that airflow accelerated over the hood and roof, reaching a maximum velocity of approximately **42.7 m/s**, while a stagnation region formed at the front bumper.
3. The pressure contour revealed high pressure at the frontal surface and a low-pressure wake behind the vehicle, indicating that pressure drag is the dominant component of the total aerodynamic drag.
4. Streamline and velocity vector analyses confirmed that flow separation occurred near the rear end of the vehicle, leading to the formation of a recirculation zone and wake vortices.
5. The turbulent kinetic energy distribution indicated that the highest turbulence intensity was concentrated in the wake region, resulting in increased aerodynamic losses.
6. Wall shear stress analysis demonstrated that frictional forces were highest near the leading edges of the vehicle and lower over the roof and rear surfaces due to flow separation.
7. The computed aerodynamic coefficients were **$C_d = 0.31$** and **$C_l = 0.08$** , which are within the typical range reported for streamlined passenger cars, validating the numerical model.
8. The residual convergence history confirmed that the simulation achieved stable convergence, ensuring the accuracy and reliability of the predicted aerodynamic parameters.
9. The study demonstrated that CFD is a cost-effective and efficient tool for evaluating vehicle aerodynamics and identifying critical flow regions before prototype development.
10. Overall, the investigation provides valuable insights into the aerodynamic performance of the scaled model car and establishes a strong foundation for future aerodynamic design improvements.

6. Future Scope

1. Future work can include wind tunnel experiments to validate the numerical results and improve the credibility of the CFD model.
2. Transient CFD simulations using advanced turbulence models such as Large Eddy Simulation (LES) or Detached Eddy Simulation (DES) can be performed to capture unsteady flow behavior more accurately.
3. The aerodynamic performance of the vehicle can be investigated at different operating speeds and Reynolds numbers to evaluate speed-dependent flow characteristics.
4. Crosswind conditions and various yaw angles can be incorporated to study vehicle stability under realistic driving environments.
5. A moving ground plane and rotating wheel models can be included to simulate actual on-road conditions and improve prediction accuracy.
6. Different vehicle body shapes, rear spoilers, diffusers, and aerodynamic accessories can be analyzed to reduce drag and improve stability.
7. Artificial intelligence and optimization techniques such as Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and Genetic Algorithms (GA) can be integrated with CFD for automated aerodynamic optimization.
8. The methodology can be extended to electric vehicles (EVs), sports cars, SUVs, and commercial vehicles to compare aerodynamic characteristics across different vehicle categories.

7. References

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